## 3.6 Summary

- **Data quality** is defined in terms of *accuracy, completeness, consistency, timeliness, believability*, and *interpretabilty*. These qualities are assessed based on the intended use of the data.
- **Data cleaning** routines attempt to fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data. Data cleaning is usually performed as an iterative two-step process consisting of discrepancy detection and data transformation.
- **Data integration** combines data from multiple sources to form a coherent data store. The resolution of semantic heterogeneity, metadata, correlation analysis, tuple duplication detection, and data conflict detection contribute to smooth data integration.
- **Data reduction** techniques obtain a reduced representation of the data while minimizing the loss of information content. These include methods of *dimensionality reduction*, *numerosity reduction*, and *data compression*. **Dimensionality reduction** reduces the number of random variables or attributes under consideration. Methods include *wavelet transforms, principal components analysis, attribute subset selection*, and *attribute creation*. **Numerosity reduction** methods use parametric or nonparatmetric models to obtain smaller representations of the original data. Parametric models store only the model parameters instead of the actual data. Examples include regression and log-linear models. Nonparamteric methods include histograms, clustering, sampling, and data cube aggregation. **Data compression** methods apply transformations to obtain a reduced or "compressed" representation of the original data. The data reduction is *lossless* if the original data can be reconstructed from the compressed data without any loss of information; otherwise, it is *lossy*.
- **Data transformation** routines convert the data into appropriate forms for mining. For example, in **normalization**, attribute data are scaled so as to fall within a small range such as 0.0 to 1.0. Other examples are **data discretization** and **concept hierarchy generation**.
- **Data discretization** transforms numeric data by mapping values to interval or concept labels. Such methods can be used to automatically generate *concept hierarchies* for the data, which allows for mining at multiple levels of granularity. Discretization techniques include binning, histogram analysis, cluster analysis, decision tree analysis, and correlation analysis. For nominal data, **concept hierarchies** may be generated based on schema definitions as well as the number of distinct values per attribute.
- Although numerous methods of data preprocessing have been developed, data preprocessing remains an active area of research, due to the huge amount of inconsistent or dirty data and the complexity of the problem.

## 3.7 Exercises

**3.1** *Data quality* can be assessed in terms of several issues, including accuracy, completeness, and consistency. For each of the above three issues, discuss how data quality assessment can depend on the *intended use* of the data, giving examples. Propose two other dimensions of data quality.

**3.2** In real-world data, tuples with *missing values* for some attributes are a common occurrence. Describe various methods for handling this problem.

**3.3** Exercise 2.2 gave the following data (in increasing order) for the attribute *age*: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

(a) Use *smoothing by bin means* to smooth these data, using a bin depth of 3. Illustrate your steps. Comment on the effect of this technique for the given data.

(b) How might you determine *outliers* in the data?

(c) What other methods are there for *data smoothing*?

**3.4** Discuss issues to consider during *data integration*.

**3.5** What are the value ranges of the following *normalization methods*?

(a) min-max normalization

(b) z-score normalization

(c) z-score normalization using the mean absolute deviation instead of standard deviation

(d) normalization by decimal scaling

**3.6** Use these methods to *normalize* the following group of data:

$$200, 300, 400, 600, 1000$$

(a) min-max normalization by setting $min = 0$ and $max = 1$

(b) z-score normalization

(c) z-score normalization using the mean absolute deviation instead of standard deviation

(d) normalization by decimal scaling

**3.7** Using the data for *age* given in Exercise 3.3, answer the following:

(a) Use min-max normalization to transform the value 35 for *age* onto the range $[0.0, 1.0]$.

(b) Use z-score normalization to transform the value 35 for *age*, where the standard deviation of *age* is 12.94 years.

(c) Use normalization by decimal scaling to transform the value 35 for *age*.

(d) Comment on which method you would prefer to use for the given data, giving reasons as to why.

**3.8** Using the data for *age* and *body fat* given in Exercise 2.4, answer the following:

(a) Normalize the two attributes based on *z-score normalization*.

(b) Calculate the *correlation coefficient* (Pearson's product moment coefficient). Are these two attributes positively or negatively correlated? Compute their covariance.

**3.9** Suppose a group of 12 *sales price* records has been sorted as follows:

$$5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215.$$

Partition them into three bins by each of the following methods:

(a) equal-frequency (equal-depth) partitioning

(b) equal-width partitioning

(c) clustering

**3.10** Use a flowchart to summarize the following procedures for *attribute subset selection*:

(a) stepwise forward selection

(b) stepwise backward elimination

(c) a combination of forward selection and backward elimination

**3.11** Using the data for *age* given in Exercise 3.3,

(a) Plot an equal-width histogram of width 10.

(b) Sketch examples of each of the following sampling techniques: SRSWOR, SRSWR, cluster sampling, and stratified sampling. Use samples of size 5 and the strata "youth," "middle-aged," and "senior."

**3.12** ChiMerge [Ker92] is a supervised, bottom-up (i.e., merge-based) *data discretization* method. It relies on $\chi^2$ analysis: Adjacent intervals with the least $\chi^2$ values are merged together until the chosen stopping criterion satisfies.

(a) Briefly describe how ChiMerge works.

(b) Take the IRIS data set, obtained from the University of California–Irvine Machine Learning Data Repository (*www.ics.uci.edu/~mlearn/MLRepository.html*), as a data set to be discretized. Perform data discretization for each of the four numeric attributes using the ChiMerge method. (Let the stopping criteria be: *max-interval* = 6). You need to write a small program to do this to avoid clumsy numerical computation. Submit your simple analysis and your test results: split-points, final intervals, and the documented source program.

**3.13** Propose an algorithm, in pseudocode or in your favorite programming language, for the following:

(a) The automatic generation of a concept hierarchy for nominal data based on the number of distinct values of attributes in the given schema.

(b) The automatic generation of a concept hierarchy for numeric data based on the *equal-width* partitioning rule.

(c) The automatic generation of a concept hierarchy for numeric data based on the *equal-frequency* partitioning rule.

**3.14** Robust data loading poses a challenge in database systems because the input data are often dirty. In many cases, an input record may miss multiple values; some records could be *contaminated*, with some data values out of range or of a different data type than expected. Work out an automated *data cleaning and loading* algorithm so that the erroneous data will be marked and contaminated data will not be mistakenly inserted into the database during data loading.

## 3.8 Bibliographic Notes

Data preprocessing is discussed in a number of textbooks, including English [Eng99], Pyle [Pyl99], Loshin [Los01], Redman [Red01], and Dasu and Johnson [DJ03]. More specific references to individual preprocessing techniques are given later.

For discussion regarding data quality, see Redman [Red92]; Wang, Storey, and Firth [WSF95]; Wand and Wang [WW96]; Ballou and Tayi [BT99]; and Olson [Ols03]. Potter's Wheel (*control.cx.berkely.edu/abc*), the interactive data cleaning tool described in Section 3.2.3, is presented in Raman and Hellerstein [RH01]. An example of the development of declarative languages for the specification of data transformation operators is given in Galhardas et al. [GFS+01]. The handling of missing attribute values is discussed in Friedman [Fri77]; Breiman, Friedman, Olshen, and Stone [BFOS84]; and Quinlan [Qui89]. Hua and Pei [HP07] presented a heuristic approach to cleaning *disguised missing data*, where such data are captured when users falsely select default values on forms (e.g., "January 1" for *birthdate*) when they do not want to disclose personal information.

A method for the detection of outlier or "garbage" patterns in a handwritten character database is given in Guyon, Matic, and Vapnik [GMV96]. Binning and data normalization are treated in many texts, including Kennedy et al. [KLV+98], Weiss and Indurkhya [WI98], and Pyle [Pyl99]. Systems that include attribute (or feature) construction include BACON by Langley, Simon, Bradshaw, and Zytkow [LSBZ87]; Stagger by Schlimmer [Sch86]; FRINGE by Pagallo [Pag89]; and AQ17-DCI by Bloedorn and Michalski [BM98]. Attribute construction is also described in Liu and Motoda [LM98a, LM98b]. Dasu et al. built a BELLMAN system and proposed a set of interesting methods for building a data quality browser by mining database structures [DJMS02].

A good survey of data reduction techniques can be found in Barbará et al. [BDF+97]. For algorithms on data cubes and their precomputation, see Sarawagi and Stonebraker [SS94]; Agarwal et al. [AAD+96]; Harinarayan, Rajaraman, and Ullman [HRU96]; Ross and Srivastava [RS97]; and Zhao, Deshpande, and Naughton [ZDN97]. Attribute subset selection (or *feature subset selection*) is described in many texts such as Neter, Kutner, Nachtsheim, and Wasserman [NKNW96]; Dash and Liu [DL97]; and Liu and Motoda [LM98a, LM98b]. A combination forward selection and backward elimination method

was proposed in Siedlecki and Sklansky [SS88]. A wrapper approach to attribute selection is described in Kohavi and John [KJ97]. Unsupervised attribute subset selection is described in Dash, Liu, and Yao [DLY97].

For a description of wavelets for dimensionality reduction, see Press, Teukolosky, Vetterling, and Flannery [PTVF07]. A general account of wavelets can be found in Hubbard [Hub96]. For a list of wavelet software packages, see Bruce, Donoho, and Gao [BDG96]. Daubechies transforms are described in Daubechies [Dau92]. The book by Press et al. [PTVF07] includes an introduction to singular value decomposition for principal components analysis. Routines for PCA are included in most statistical software packages such as SAS (*www.sas.com/SASHome.html*).

An introduction to regression and log-linear models can be found in several textbooks such as James [Jam85]; Dobson [Dob90]; Johnson and Wichern [JW92]; Devore [Dev95]; and Neter, Kutner, Nachtsheim, and Wasserman [NKNW96]. For log-linear models (known as *multiplicative models* in the computer science literature), see Pearl [Pea88]. For a general introduction to histograms, see Barbará et al. [BDF+97] and Devore and Peck [DP97]. For extensions of single-attribute histograms to multiple attributes, see Muralikrishna and DeWitt [MD88] and Poosala and Ioannidis [PI97]. Several references to clustering algorithms are given in Chapters 10 and 11 of this book, which are devoted to the topic.

A survey of multidimensional indexing structures is given in Gaede and Günther [GG98]. The use of multidimensional index trees for data aggregation is discussed in Aoki [Aok98]. Index trees include R-trees (Guttman [Gut84]), quad-trees (Finkel and Bentley [FB74]), and their variations. For discussion on sampling and data mining, see Kivinen and Mannila [KM94] and John and Langley [JL96].

There are many methods for assessing attribute relevance. Each has its own bias. The information gain measure is biased toward attributes with many values. Many alternatives have been proposed, such as gain ratio (Quinlan [Qui93]), which considers the probability of each attribute value. Other relevance measures include the Gini index (Breiman, Friedman, Olshen, and Stone [BFOS84]), the $\chi^2$ contingency table statistic, and the uncertainty coefficient (Johnson and Wichern [JW92]). For a comparison of attribute selection measures for decision tree induction, see Buntine and Niblett [BN92]. For additional methods, see Liu and Motoda [LM98a], Dash and Liu [DL97], and Almuallim and Dietterich [AD91].

Liu et al. [LHTD02] performed a comprehensive survey of data discretization methods. Entropy-based discretization with the C4.5 algorithm is described in Quinlan [Qui93]. In Catlett [Cat91], the D-2 system binarizes a numeric feature recursively. ChiMerge by Kerber [Ker92] and Chi2 by Liu and Setiono [LS95] are methods for the automatic discretization of numeric attributes that both employ the $\chi^2$ statistic. Fayyad and Irani [FI93] apply the minimum description length principle to determine the number of intervals for numeric discretization. Concept hierarchies and their automatic generation from categorical data are described in Han and Fu [HF94].

# 4 Data Warehousing and Online Analytical Processing

**Data warehouses generalize** and consolidate data in multidimensional space. The construction of data warehouses involves data cleaning, data integration, and data transformation, and can be viewed as an important preprocessing step for data mining. Moreover, data warehouses provide *online analytical processing (OLAP)* tools for the interactive analysis of multidimensional data of varied granularities, which facilitates effective data generalization and data mining. Many other data mining functions, such as association, classification, prediction, and clustering, can be integrated with OLAP operations to enhance interactive mining of knowledge at multiple levels of abstraction. Hence, the data warehouse has become an increasingly important platform for data analysis and OLAP and will provide an effective platform for data mining. Therefore, data warehousing and OLAP form an essential step in the knowledge discovery process. This chapter presents an overview of data warehouse and OLAP technology. This overview is essential for understanding the overall data mining and knowledge discovery process.

In this chapter, we study a well-accepted definition of the data warehouse and see why more and more organizations are building data warehouses for the analysis of their data (Section 4.1). In particular, we study the *data cube*, a multidimensional data model for data warehouses and OLAP, as well as OLAP operations such as roll-up, drill-down, slicing, and dicing (Section 4.2). We also look at data warehouse design and usage (Section 4.3). In addition, we discuss *multidimensional data mining*, a powerful paradigm that integrates data warehouse and OLAP technology with that of data mining. An overview of data warehouse implementation examines general strategies for efficient data cube computation, OLAP data indexing, and OLAP query processing (Section 4.4). Finally, we study data generalization by attribute-oriented induction (Section 4.5). This method uses concept hierarchies to generalize data to multiple levels of abstraction.

## 4.1 Data Warehouse: Basic Concepts

This section gives an introduction to data warehouses. We begin with a definition of the data warehouse (Section 4.1.1). We outline the differences between operational database

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